

Lab6

Continuous Integration with Jenkins: Setting Up a CI Pipeline, Integrating Jenkins with Maven/Gradle, Running Automated Builds and Tests

Step 1: Login with user name and password

Step 2: Manage Jenkins-Tools-select maven installation->name->MAVEN_HOME->deselect the install automatically->copy the path of maven in your system and paste and save.

Step 3: Go to plugins->available plugins->search for maven integration->maven integration 3.2X and click on install.

Step 4: dashboard->new item->program6->click on freestyle project and ok.

Step 5: build set up->click on add build set up->select invoke top level maven targets->maven version select maven_home

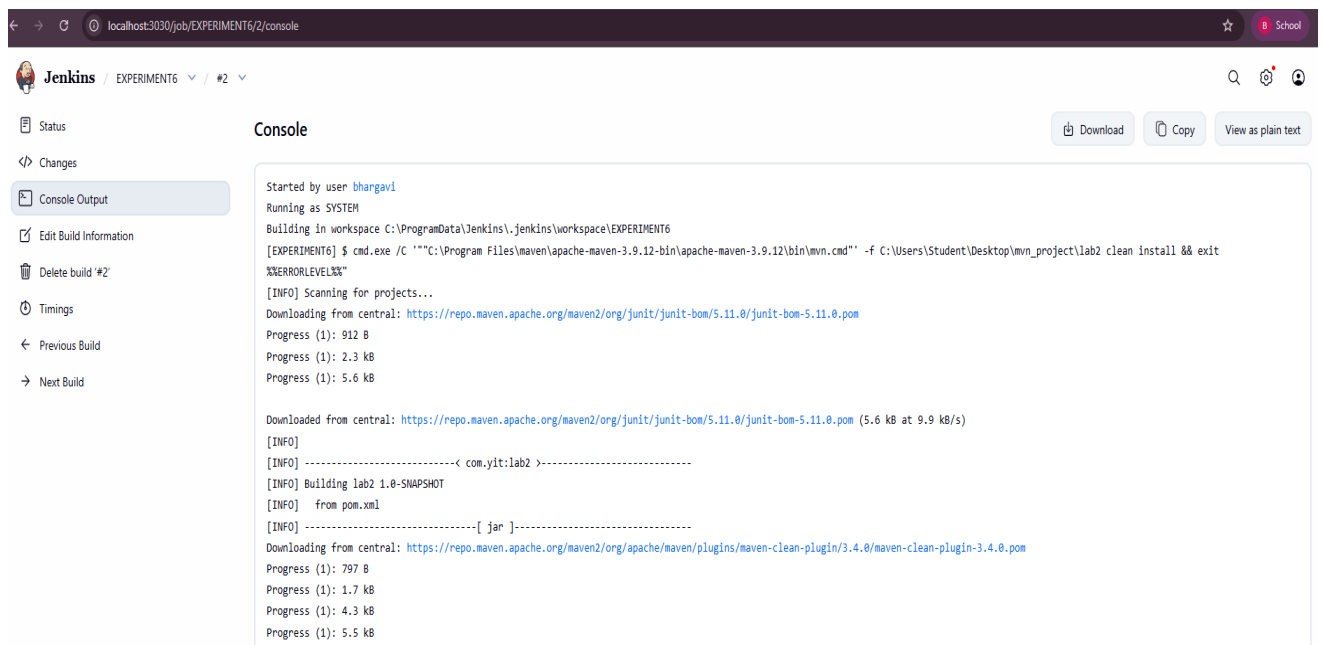
Goal->type the following

Clean install

Click on advance->pom->copy the path of pom.xml of program2 and paste, save.

Step 6: build now

Step 7: goto console and check the result



The screenshot shows the Jenkins web interface for a build named 'EXPERIMENT6' (build #2). The 'Console' tab is selected, displaying the build's output. The output shows the build was started by user 'bhargavi' and is running as 'SYSTEM'. It details the execution of 'mvn clean install' in a workspace. The console output includes progress bars for downloading dependencies from the central repository, such as 'junit-junit-bom-5.11.0.jar' and 'maven-clean-plugin-3.4.0.jar'. The build is currently in the 'Building' phase, with the next step being 'Building lab2 1.0-SNAPSHOT'.

```
Started by user bhargavi
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\EXPERIMENT6
[EXPERIMENT6] $ cmd.exe /C ""C:\Program Files\maven\apache-maven-3.9.12\bin\apache-maven-3.9.12\bin\mvn.cmd"" -f C:\Users\Student\Desktop\mvn_project\lab2 clean install && exit
%ERRORLEVEL%
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.11.0/junit-bom-5.11.0.pom
Progress (1): 912 B
Progress (1): 2.3 kB
Progress (1): 5.6 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.11.0/junit-bom-5.11.0.pom (5.6 kB at 9.9 kB/s)
[INFO]
[INFO] -----< com.yit:lab2 >-----
[INFO] Building lab2 1.0-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.4.0/maven-clean-plugin-3.4.0.pom
Progress (1): 797 B
Progress (1): 1.7 kB
Progress (1): 4.3 kB
Progress (1): 5.5 kB
```

```
Jenkins / EXPERIMENT6 / #2

Progress (1): 94/157 kB
Progress (1): 111/157 kB
Progress (1): 127/157 kB
Progress (2): 127/157 kB | 7.7/196 kB
Progress (2): 143/157 kB | 7.7/196 kB
Progress (2): 143/157 kB | 16/196 kB
Progress (2): 157 kB | 16/196 kB
Progress (2): 157 kB | 32/196 kB
Progress (2): 157 kB | 48/196 kB
Progress (2): 157 kB | 65/196 kB
Progress (2): 157 kB | 81/196 kB
Progress (2): 157 kB | 98/196 kB
Progress (2): 157 kB | 114/196 kB
Progress (2): 157 kB | 130/196 kB
Progress (2): 157 kB | 147/196 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 1.6 MB/s)
Progress (1): 163/196 kB
Progress (1): 179/196 kB
Progress (1): 196 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 899 kB/s)
[INFO] Installing C:\Users\Student\Desktop\mvn_project\lab2\pom.xml to C:\Windows\system32\config\systemprofile\.m2\repository\com\yit\lab2\1.0-SNAPSHOT\lab2-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\Student\Desktop\mvn_project\lab2\target\lab2-1.0-SNAPSHOT.jar to C:\Windows\system32\config\systemprofile\.m2\repository\com\yit\lab2\1.0-SNAPSHOT\lab2-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 26.738 s
[INFO] Finished at: 2026-03-10T11:43:27+05:30
[INFO] -----
Finished: SUCCESS
```

To test

Step 1: go to build history click on configure change the goal to test and save build again

```
localhost:3030/job/EXPERIMENT6/3/console

Jenkins / EXPERIMENT6 / #3

Status
Changes
Console Output
Edit Build Information
Delete build #3
Timings
Previous Build

Console

Started by user bhargavi
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\EXPERIMENT6
[EXPERIMENT6] $ cmd.exe /C ""C:\Program Files\maven\apache-maven-3.9.12\bin\mvn.cmd"" -f C:\Users\Student\Desktop\mvn_project\lab2 test && exit %ERRORLEVEL%
[INFO] Scanning for projects...
[INFO] -----< com.yit:lab2 >-----
[INFO] Building lab2 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.3.1:resources (default-resources) @ lab2 ---
[INFO] skip non existing resourceDirectory C:\Users\Student\Desktop\mvn_project\lab2\src\main\resources
[INFO] --- compiler:3.13.0:compile (default-compile) @ lab2 ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.3.1:testResources (default-testResources) @ lab2 ---
[INFO] skip non existing resourceDirectory C:\Users\Student\Desktop\mvn_project\lab2\src\test\resources
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ lab2 ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- surefire:3.3.0:test (default-test) @ lab2 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.yit.demo.AppTest
Running test: 2 + 3 = 5
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.050 s -- in com.yit.demo.AppTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.407 s
[INFO] Finished at: 2026-03-10T11:45:40+05:30
[INFO] -----
Finished: SUCCESS
```

